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Axisymmetric Flow Fields**

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VISCOUS OPTIMIZED HYPERSONIC WAVERIDERS DESIGNED FROM AXISYMMETRIC FLOW FIELDS

by

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ABSTRACT

A series of "viscous optimized" waveriders is designed from general axisymmetric flow fields, other than conical -- most notably, flow fields over minimum drag bodies. These configurations represent the next logical step in an ongoing program of research on the design of hypersonic waveriders at the University of Maryland. In previous work at Maryland, inviscid conical flow was used exclusively in the design of waveriders. These conical flow waveriders predict high values of lift/drag, higher than the values based on experience from other hypersonic configurations. The present work allows the design of hypersonic waveriders from any axisymmetric flow field -- although the present results focus on the inviscid flows over cones and 3/4 and 1/2 power-law bodies. A space marching, finite-difference code is used to generate the axisymmetric flow field. The undersurface of the vehicle is carved out as a stream surface of this axisymmetric flow field, whereas the upper surface is assumed to be a freestream surface. The detailed viscous effects are included within the optimization process, using a simple reference temperature method. Results obtained using the reference temperature method are within 10% of results obtained using a more complex integral boundary layer method, even at high hypersonic Mach numbers. Boundary layer transition is predicted using a correlation of the local transition Reynolds number with the Mach number. A non-linear simplex method is used to optimize the waveriders for either maximum L/D or minimum total drag.

INTRODUCTION

In recent years, research involving all aspects of hypersonic flight has seen a rebirth in the United States. An extended survey of hypersonic aerodynamic research is given in Reference 1. During the long hiatus that hypersonic research lay dormant, powerful new engineering tools have become available -- i.e. supercomputers and computational fluid dynamics (CFD). In fact, vehicle concepts such as the National Aerospace Plane (NASP), aero-assisted orbital transfer

vehicles (AOTV's), the hypersonic transport (the "Orient Express"), and hypersonic missiles, place much of the design burden upon CFD. In addition, advanced computers and computational techniques allow us to take a "new" look at some "old" hypersonic aerodynamic concepts. The present paper is part of a continuing hypersonics research program, at the University of Maryland, that does just that. Using modern computational techniques, the "old" concept of waveriders is revisited to produce a "new" class of advanced hypersonic lifting configurations. This CFD approach to the waverider concept has lead to a design tool with the following "new" advantages:

1. The waverider configurations can be numerically optimized for almost any figure of merit -- maximum L/D and minimum total drag, in the present study.
2. The detailed viscous effects are included within the optimization process.
3. The waverider configurations can be carved out as stream surfaces of almost any axisymmetric flow field -- the inviscid flow over cones and 3/4 and 1/2 power-law bodies are used in the present investigation.
4. Even with the above detailed effects and versatility, the waverider optimization procedure is computationally efficient, allowing parametric design studies to be made.

The advantages listed above will be elaborated upon in the paragraphs to come.

This paper is a sequel to Reference 2. In References 2 and 3, Bowcutt and Anderson established a new class of hypersonic vehicles that broke the "L/D barrier" as depicted in Figure 1. This barrier represents a general empirical correlation for $(L/D)_{max}$, as the freestream Mach number, M_∞ , increases across the supersonic and hypersonic regimes, based on actual flight vehicle experience. It is given by the relation

$$(L/D)_{max} = \frac{4(M_\infty + 3)}{M_\infty}$$

(obtained from Reference 4), and is shown by the solid line in Figure 1. A number of data points

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for optimized waveriders generated by Bowcutt and Corda are shown along with various other hypersonic configurations at various Reynolds numbers and Mach numbers. (The points on this graph correspond to different references as given in References 5 and 6.) The waveriders generated by Bowcutt are seen to follow the relation given by

$$(L/D)_{\max} = \frac{5(M_{\infty} + 2)}{M_{\infty}}$$

shown by the dashed curve in Figure 1. The present waverider results, as represented by the solid circles in Figure 1, will be discussed in subsequent sections.

To fully appreciate the logical sequence of the present waverider research and how it represents a contribution to the state-of-the-art, let us review the fundamental concepts of waveriders. In 1959, Nonweiler⁷ proposed the idea of constructing a three-dimensional hypersonic vehicle from a known flow field. Nonweiler chose the flow field behind a planar oblique shock wave to generate a class of vehicles with a caret-shaped cross-section and a delta planform. The construction of a caret wing, as they are called, is shown in Figure 2 (from Reference 8). When flying at its design Mach number, the body appears to be riding on top of the attached shock wave, and hence is dubbed a "waverider". There is no flow spillage from the lower to the upper surface of the vehicle when it is flying at its design Mach number because the shock wave is attached to the leading edges. This containment of the flow beneath the vehicle results in high pressure being exerted on the lower surface -- the high pressure behind the two-dimensional planar shock wave. This high pressure leads to high lift. It can be stated that, in general, when compared at the same lift coefficient, caret wings have higher values of L/D than other winged hypersonic vehicles. The aerodynamic benefits of waveriders are listed in Reference 1, and are discussed in detail in References 5, 6, and 9.

The idea of generating a three-dimensional hypersonic vehicle from a known flow field can be extended from Nonweiler's construction (which uses a simple two-dimensional wedge) to more complex flows. For example, a conical flow can be used, as shown in Figure 3 (from Reference 8). The resulting class of "conical-flow" waveriders was first investigated by the British (see Reference 10). Rasmussen et al.^{11,12}, Cole and Zien¹³, and Kim et al.¹⁴ represent further variations -- designing waveriders from flows over cones and elliptic cylinders using hypersonic small disturbance theory. Further, these investigators used the calculus of variations to search for optimum waverider shapes. Bowcutt and Anderson², generated a series of conical-flow waveriders that were numerically optimized for maximum L/D. Unlike previous investigators, they are the first to include the detailed viscous effects (including boundary layer transition) within the optimization process. This helps to alleviate a fundamental drawback of waveriders -- the large friction drag associated with a waverider's characteristically large wetted area. With skin friction effects fully accounted for within the optimization

procedure, a configuration can be found which balances the desire for aerodynamic efficiency (high L/D) with the large wetted area (hence high friction drag) attendant with this high L/D.

The present work takes the next logical step, going beyond conical flow fields, to generate "viscous optimized" waveriders from general axisymmetric flow fields -- most notably, flow over minimum drag bodies. Again, the detailed viscous effects are included within the optimization process. The rationale behind this direction is as follows. One might intuitively suspect that the flow over a minimum drag power-law body might lead to waveriders of lower drag and higher L/D than those derived from conical flow, by nature of the generating flow field itself. The need for minimum drag and high L/D are of paramount importance for vehicles (such as the aerospace plane) that must be able to accelerate and cruise as efficiently as possible. One might also expect a better volume distribution (meaning more volume in the nose of the vehicle for packaging of payload or avionics) with a power-law derived waverider than with a conically derived waverider. In particular, the present results show waveriders designed from the flow over cones and 1/2 and 3/4 power-law bodies. Also, the configurations are numerically optimized, using the non-linear simplex method of Nelder and Mead¹⁵, for maximum L/D and for minimum total drag.

ANALYSIS

The analysis and optimization of waveriders from general axisymmetric flow fields require distinct capabilities. First, one must be able to generate a "known" axisymmetric flow field. Then one must analyze the waverider generated from the flow field which involves: 1) the evaluation of the aerodynamic coefficients (e.g. C_L , C_D , and C_M) and 2) the determination of the parameters which describe the waverider geometry. Finally, waverider optimization requires finding the set of parameters that yield the desired figure of merit (e.g. maximum L/D or minimum C_D). The breakdown of the waverider design process is shown in Figure 4. The contents of each box will be discussed in the following paragraphs. A more detailed description of the design process is given in Reference 6.

A. Generation of Inviscid Flow Field

The calculation of the "known" axisymmetric flow field will be discussed first. Separate techniques are used for the case of 1) flow over a right circular cone and 2) flow over a general axisymmetric body, as described below.

A.1 Conical Flow

For the case of conical flow, the ordinary differential equation known as the Taylor-Maccoll equation is solved using a standard fourth-order Runge-Kutta method as given in Reference 16 and shown below.

$$\frac{\gamma-1}{2} [1-V_r^2 - (\frac{dV}{d\theta} r)^2] [2V_r + \frac{dV}{d\theta} r \cot \theta + \frac{d^2 V}{d\theta^2} r] - \frac{dV}{d\theta} r [V_r' \frac{dV}{d\theta} r + \frac{dV}{d\theta} r \frac{d^2 V}{d\theta^2} r] = 0 \quad (1)$$

In equation (1), V_r is the nondimensional component of flow velocity along a conical ray, θ is the angle of the ray referenced to the cone axis, and γ is the ratio of specific heats.

A2. General Axisymmetric Flow Field

For an inviscid axisymmetric flow field in general, the governing partial differential equations are the Euler equations given below

$$\frac{\partial G}{\partial z} = - \frac{\partial E}{\partial r} - H \quad (2)$$

$$G = \begin{bmatrix} \rho w \\ p + \rho w^2 \\ \rho vw \end{bmatrix}$$

$$E = \begin{bmatrix} \rho v \\ \rho vw \\ p + \rho v^2 \end{bmatrix}$$

$$H = \frac{1}{r} \begin{bmatrix} \rho v \\ \rho vw \\ \rho v^2 \end{bmatrix}$$

where ρ is the density, w is the axial (z) component of the flow velocity, v is the transverse (r) component of the flow velocity, and p is the pressure.

These equations are solved numerically using MacCormack's explicit, space-marching, finite-difference scheme. In this technique, the flow field of interest is discretized into a network of grid points in the axial (z) and radial (r) directions. Starting from an initial data plane, where the flow properties are assumed known, the flow field solution is calculated by marching downstream in steps of Δz . The shock-fitted approach is used where the solution is bounded by the surface of the body and the shock wave produced by the body. Applied to the Euler equations, the scheme can be written as

$$\overline{G_j^{n+1}} = G_j^n - \frac{\Delta z}{\Delta r} (E_{j+1}^n - E_j^n) - \Delta z H_j^n \quad (3a)$$

$$\overline{G_j^{n+1}} = \frac{1}{2} \overline{G_j^n} + \overline{G_j^{n+1}} - \frac{\Delta z}{\Delta r} (\overline{E_j^{n+1}} - \overline{E_{j-1}^{n+1}}) - \Delta z \overline{H_j^{n+1}} \quad (3b)$$

where Δz is the marching increment in the axial (z) direction, Δr is the distance between grid points in the radial (r) direction, n is the grid point index in the axial direction, and j is the grid point index in the radial direction. To obtain a stable solution, the downstream marching is limited by the CFL criterion

$$\Delta z = C \frac{\Delta r}{\tan(\theta + \mu)} \quad (4)$$

where C is a constant between 0 and 1. A value of C equal to 0.95 is used in all of the calculations presented in this paper.

As mentioned earlier, the flow field solution is started from a known data line. For this reason, the body of interest is assumed to have a conical nosetip. The flow field behind the conical nosetip is solved using the Taylor-Maccoll cone flow solution outlined in the previous section. This nosetip flow field is then matched with the body of interest to serve as an initial data line for the space-marching solution. For a detailed account of MacCormack's space-marching technique, including the application of the boundary conditions, see References 6 or 17. body of interest to serve as an initial data line for the space-marching solution. For a detailed account of MacCormack's space-marching technique, including the application of the boundary conditions, see References 6 or 17.

B. Generation of Leading Edge Shapes

Once the flow field is calculated, waverider shapes can be "carved" out of the flow field as shown in Figure 5. As is seen in this figure, the leading edge curve is traced back along the streamlines of the flow field to generate the lower surface of a waverider. Note that the projection of the leading edge curve in the cross-stream (x - y) plane uniquely defines a waverider geometry for a given flow field. The leading edge curves are generated by the optimization process as will be discussed shortly.

C. Streamline Tracing

As noted in the previous section, streamlines are traced from the leading edge, through the "known" flow field, to create the waverider lower surface. The manner of this streamline tracing will now be described.

From the space-marching solution, flow properties are known at distinct axial (z) stations. Tracing a streamline amounts to finding where the value of the streamfunction, ψ , along each z data line is equal to the value of the streamfunction of the specified streamline. The streamfunction along each data line can be calculated by using the definition of the streamfunction for an axially symmetric flow.

$$\psi_{j+1} = \psi_j + \dot{m} \quad (5)$$

where ψ is the value of the streamfunction, j is a radial (r) index for data points along a z station, and $\dot{m}$ is the mass flow in the annulus bounded by the points j and $j+1$. By setting the value of the streamfunction at the body surface equal to zero, the value of the streamfunction can be calculated along each z data line.

D. Freestream Upper Surface

The generation of the waverider lower surface by streamline tracing has been described in the previous sections. The upper surface is created by simply following the freestream back through the given leading edge curve to the base of the waverider. The pressure on this freestream upper surface is the freestream pressure, p_∞ .

E. Skin Friction Calculations

The skin friction distribution along the streamlines that form the waverider is calculated using the reference temperature method of Eckert¹⁸. In the reference temperature method, approximate formulas are used to predict the skin friction, with the physical properties evaluated at an appropriate reference temperature. For a flat plate in laminar flow, the local skin friction coefficient is given by

$$c_f = 0.664 (Re_x)^{-1/2} (T'/T_\infty)^{(\omega-1)/2} \quad (6)$$

Re_x is the local Reynolds number defined as

$$Re_x = \frac{\rho_\infty V_\infty x}{\mu_\infty} \quad (7)$$

where ρ_∞ is the freestream density, V_∞ is the freestream velocity, x is the local distance from the leading edge of the plate, and μ_∞ is the freestream value of the viscosity.

Also in equation (6), T' is the reference temperature defined as

$$(T'/T_\infty) = 1 + 0.032 M_\infty^2 + 0.58 (T_w/T_\infty - 1) \quad (8)$$

where M_∞ is the freestream Mach number and T_w is the wall temperature. Finally, ω in equation (6) is the exponent of an assumed exponential variation of μ , namely,

$$(\mu'/\mu_\infty) = (T'/T_\infty)^\omega \quad (9)$$

A value of $\omega = 0.75$ is used in the present study.

The flat plate skin friction coefficient for turbulent flow is given by

$$c_f = \frac{0.0592}{(Re'_x)^{0.2}} \quad (10a)$$

where

$$Re'_x = \frac{\rho' V_\infty x}{\mu'} \quad (10b)$$

In equation (10b), ρ' and μ' are evaluated at the reference temperature T' .

This type of skin friction analysis is much simpler, in concept and use, than integral boundary layer methods (such as used in Reference 2). Results obtained using the reference temperature method were within 10% of results using a more complex integral boundary layer method (Reference 2), even at high hypersonic Mach numbers. Also, the computation time required by the reference temperature method is very small when compared with an integral boundary layer method.

Boundary layer transition is predicted using a correlation of the local transition Reynolds number, $Re_{x,t}$ with the local edge Mach number, Me , as follows

$$\log_{10} Re_{x,t} = 6.421 e^{(1.209 \times 10^{-4} Me^{2.641})} \quad (11)$$

This correlation is the same as used in the previous work by Bowcutt and Anderson^{2,3} and is based on the experimental data of DiCristina¹⁹ for transition on sharp cones at zero degrees angle of attack. This correlation is used due to the lack of better methods of transition prediction in hypersonic flows.

F. Aerodynamic Forces and Moments

The aerodynamic force and moment coefficients are calculated by numerically integrating the pressure and shear stress over the surface of the waverider. Base drag is not calculated. Details of the integration for the forces and moments can be found in Reference 6.

G. Waverider Optimization

The construction and aerodynamic analysis of a single waverider configuration is shown by blocks A through F, in Figure 4, and just described in the previous paragraphs. Now, the optimization is performed by perturbing the shape of the leading edge curve (which corresponds to a unique waverider geometry), until a configuration is found with the optimum value of the specified figure of merit -- either maximum L/D or minimum total drag, in the present analysis. The non-linear simplex method of Nelder and Mead¹⁵ is used for the numerical optimization. Because this is a zero order method, only function evaluations are needed to find an optimum (no derivative information is needed). These function evaluations are the determinations of the figure of merit (e.g. L/D or C_D) for each configuration. The zero order methods are the easiest to implement and it was shown by VanWie²⁰ that the non-linear simplex method worked as well as other higher-order methods when applied to a similar type of optimization problem as is considered here.

The non-linear simplex method minimizes a function of n variables by comparing values of the function at $(n+1)$ vertices of a "simplex", replacing the vertex with the highest function value with another point determined by the logic of the scheme, and then "moving" this simplex, over the function surface, in the direction of the function minimum. Three operations are used by

the simplex in its search for the function minimum -- reflection, expansion, and contraction. These operations are graphically illustrated by considering the minimization problem shown in Figure 6. Here, the minimum of a function of two variables (x_1 and x_2) is sought, where the contour map defines the function. A simplex composed of $n+1=3$ vertices (a triangle) moves over the function surface by reflecting or "flip-flopping" down the function valley, expanding if possible to speed up the process, and finally contracting around the function minimum. Constraints in the optimization process take the form of "barriers" or boundaries that the simplex is not allowed to cross. The nature of these constraints will be discussed in a later section.

In the present work, the function to be minimized is the negative of the lift/drag ratio or the value of the total drag coefficient. Each function evaluation corresponds to a unique waverider configuration, which can be parameterized by its leading edge shape. Therefore, a leading edge curve represents a single vertex point of the simplex. The leading edge is specified by five points in the cross-stream (x - y) plane, where a cubic spline is used to define a smooth, continuous curve. It should be noted that calculations are performed for only half of the vehicle because of its longitudinal symmetry. One of the five points is constrained to lie on the symmetry plane (y -axis) of the waverider, thus reducing the number of x , y coordinates, needed to define the leading edge, from 10 to 9. The number of coordinate variables is further reduced to 8 by the constraint that the last leading edge point must lie on the shock wave. It is these eight variables that are perturbed by the logic of the optimization routine to find an optimum waverider.

Since the optimization problem is reduced to finding the minimum of a function defined by eight variables ($n=8$), the simplex for this problem has $n+1$ or nine vertices with each vertex corresponding to a set of coordinates for a leading edge curve. To initiate the simplex method, nine leading edge shapes must be chosen -- so-called basis leading edge shapes. In the present study, six of the basis leading edge shapes are polynomials of the form

$$y_{1e} = C_1 + C_2 x_{1e} + C_3 x_{1e}^2 + C_4 x_{1e}^3 \quad (12)$$

and three of the basis shapes are of the forms

$$y_{1e} = C_5 + C_6 \left(1 - \cos \frac{\pi x_{1e}}{r_s} \right) \quad (13a)$$

$$y_{1e} = C_7 + C_8 \sin \left(\frac{\pi x_{1e}}{r_s} \right) \quad (13b)$$

where x_{1e} and y_{1e} are the x and y coordinates of the leading edge and r_s is the radius of the shock wave at the base of the waverider. A set of basis leading edge shapes is shown in Figure 7. The final optimized leading edge curve is also shown as the bold line in this figure. A typical optimization history is shown in Figure 8. In all of the results presented, 100 optimization steps or

levels are used to reach an "optimum" waverider configuration.

III. RESULTS AND DISCUSSION

The presented results are representative of the work done in Reference 6. The following areas will be covered: (1) validation of the computer code used to construct and aerodynamically analyze a waverider (no optimization); (2) a presentation of optimum waveriders designed from conical and power-law body flow fields, over a wide range of hypersonic Mach numbers; (3) the comparison of waveriders optimized for maximum L/D with those optimized for minimum total drag; (4) the effect of using an all laminar, all turbulent, or transiting boundary layer in the waverider optimization; and (5) a word will be said concerning the computing time required.

A. Code Validation

Because of the unique design method used in the present analysis, it is impossible to find experimental data to confirm the computational results presented. However, certain point comparisons can be made. One such case is the hypersonic flow over a half-cone underneath a delta wing configuration at zero degrees angle of attack. Experiments by Fetterman²¹ on a 4 degree semi-apex angle half-cone under a delta wing of 81 degrees sweepback closely approximate tests of a waverider configuration at a Mach number of 6.86. Table 1 summarizes the results of this comparison. The wall temperature of the model in the experiments is unknown, so calculations are made for a wall temperature equal to the freestream temperature and equal to the total temperature of the flow. Also, the model used in the experiment has a 1 degree compression on the upper surface of the delta wing from the leading edge to approximately the midchord position. It was not feasible to model this compression surface so calculations were made for a freestream upper surface and for a 1 degree compression on the entire upper surface. Table 1 shows that calculations of the lift coefficient, C_L , are in good agreement with experiment. The calculated values of the drag coefficient, C_D , for the freestream and 1 degree compression upper surface, bracket the experimental value as would be expected. This same trend follows through to the comparison of the lift/drag ratio.

Another source for comparison of the present results is the previous work by Bowcutt², for conical flows only. This is possible because all of the computational tools used in the present investigation were developed independently of this previous work. Indeed, for the case of conical flow waveriders, the present code and the previous code developed by Bowcutt differ by a few percent for most parameters of interest. For a Mach number equal to 6 and a cone angle of 12 degrees, the present code predicts an L/D of 7.34 and Bowcutt's code predicts an L/D of 7.73 -- a difference of about 5 percent. For a higher Mach number of 14, the values of the lift/drag ratio are 6.06 and 5.87 from Bowcutt's code and the present code respectively -- a difference of about 3 percent. This comparison also helps to validate the reference temperature method, used for the

viscous flow analysis in the present work, since Bowcutt's code used a much more complex integral boundary layer method. The comparison is especially encouraging at the higher Mach numbers where the accuracy of the reference temperature method may be suspect.

8. Waveriders Designed From Cones and Power-Law Bodies

Waveriders are optimized at specific design points (e.g. Mach number and altitude) along the possible flight trajectory of an aerospace plane. These design points are flight Mach numbers of 4, 6, 10, 14, and 20 with corresponding altitudes of 80,000, 110,000, 125,000, 140,000, and 175,000 feet. All calculations are for a calorically perfect gas.

The design of the waveriders are optimized under a set of geometric constraints that are easily specified by the program user. These constraints include the following:

1. A constraint on the lower limit of the slenderness ratio (reciprocal of the fineness ratio) of the waverider--equal to 0.075 times the length of the vehicle, in the present results.
2. Constraints on the box size (semi-span-to-length ratio) of the waverider--the lower limit is 0.1 and the upper limit is 0.4, in the present study.
3. A constraint on the minimum acceptable total volume of the vehicle -- not an active constraint for the present results.

A study was made to assess the effect of these constraints on the design process but will not be discussed here. Reference 6 can be consulted for complete details.

The present results are for waveriders designed from flow over cones and 1/2 and 3/4 power-law bodies. Typical body shapes are illustrated in Figure 9 for a Mach number of 10 and a base radius-to-length ratio, r_{base}/L , of 0.15. For a given Mach number and body type, the design strategy is as follows. A fineness ratio of the flow generating body is selected and the flow field corresponding to this body is calculated using either the Taylor-Maccoll solution (if the body is a cone) or the space-marching technique (if the body is a power-law body). The optimum waverider is then found for this flow field. Additional fineness ratios are selected and the optimum waverider corresponding to each of these flow fields is obtained. Figure 10 shows a series of waveriders optimized for maximum L/D for a Mach number of 6. Each one of the data points corresponds to a waverider that is optimized for a given fineness ratio of the flow field generating body, although the lift/drag ratio is plotted versus the fineness ratio of the waverider itself. From Figure 10, it is seen that there is an "optimum of the optimums" for each of the body types. In general, it is also evident that, for a given waverider fineness ratio, the waveriders designed from the 1/2 power-law body have the highest L/D, followed by

those designed from the 3/4 power-law body and those designed from the cone.

All of the remaining results in this section are for waveriders that are optimized for maximum L/D. Figure 11 shows a plot of the lift/drag ratio versus the Mach number for waveriders that are the "optimum of the optimums". In general, the power-law derived waveriders have slightly higher values of L/D than the conically derived waveriders over the Mach number range shown. The waveriders designed from 1/2 power-law bodies are seen to be the best overall, in terms of high L/D. Also note that, for all the body types, the value of L/D decreases then increases with increasing Mach number. The waverider configurations designed from the 1/2 power-law body and the cone, for Mach numbers of 4 and 20, are shown in Figures 12 and 13.

Figure 14 shows a plot of the lift/drag ratio versus the lift coefficient, C_L , for the optimum waveriders discussed in the previous paragraph. Note that decreasing C_L in this figure corresponds to an increasing Mach number. Here, the power-law body derived waveriders generally predict higher values of L/D, for a given value of C_L , than the cone-derived waveriders. Again, note the decrease then increase in L/D as the value of C_L decreases (Mach number increasing).

Figure 15 shows L/D plotted against a Mach number times the fineness ratio of the flow generating body. This figure shows a reversal in trends, in terms of which flow field yields the waverider with the highest L/D. The power-law bodies are better at the lower values of $M_\infty(b/l_b)$, and the cones are better at higher values. Also, it is seen that the shapes of the curves in Figure 15 are similar for each type of flow field. This tends to suggest that the quantity $M_\infty(b/l_b)$ is a valid hypersonic similarity parameter for the type of waverider design process presented in this paper.

Returning to figure 1, note the symbols denoted by solid circles. These symbols correspond to the waveriders with the highest L/D, designed from the flow over 1/2 power-law bodies, at the various Mach numbers. Note that they also break the "L/D barrier" as described earlier. As described earlier, the data for L/D displays a "bucket" type behavior with increasing Mach number. The decrease then increase in the value of L/D, as Mach number increases, is a Reynolds number effect as discussed in Reference 2.

The total drag coefficient, C_D , is plotted against Mach number for the "optimum of the optimums" waveriders in Figure 16. The value of drag coefficient is seen to decrease with increasing Mach number for all of the flow types. Note that, except for the Mach 6 case, the cone-flow derived waveriders have the slightly higher values of drag coefficient. It is interesting to note, in Figure 17, the breakdown of drag, into wave and skin friction drag, for the 3/4 power-law body flow. It is seen that the skin friction drag and wave drag are nearly equal throughout the Mach number regime. These trends are true for the other flow field results.

Finally, Figure 18 shows volumetric efficiency, $V^{2/3}/S$ (where V is the total volume of the waverider and S is its planform area) versus Mach number for the optimum waveriders. This figure is shown to indicate the order of magnitude of the volumetric efficiency for the present class of waverider vehicles. In general, the power-law body derived waveriders appear to be more volume efficient than their conically derived counterparts, but the trend is not consistent.

C. Maximum L/D Versus Minimum Total Drag

This section compares the waveriders that are optimized for maximum L/D with those optimized for minimum total drag. Maximizing for L/D can be thought of as searching for the most efficient hypersonic cruising configuration, while minimizing C_D corresponds to finding the best accelerator type hypersonic vehicle. Figure 19 compares the configurations of two cone flow derived waveriders at Mach 6 for these two types of optimization. The same comparison is made for waveriders designed from flow over a 3/4 power-law body, at a Mach number of 14, in Figure 20. It is evident that optimizing for maximum L/D and minimizing for total drag lead to quite different waverider configurations. For the cone-flow Mach 6 case shown in Figure 19, the value of L/D is 7.175 when maximizing L/D and 7.187 when minimizing drag. The value of the drag coefficient, C_D , for the drag minimization case ($C_D = 0.004222$) is lower than for the case of maximizing L/D ($C_D = 0.004694$). The corresponding numbers for the Mach 14, power-law body case are L/D equal to 6.484 and C_D of 0.003095 for the maximization of L/D and L/D equal to 6.533 and C_D of 0.003033 for the total drag minimization. These are the usual trends seen in the data for all of the cases run.

D. Study Concerning the Nature of the Boundary Layer

A study was made to determine the effect of using an all laminar, an all turbulent, and a transiting boundary layer in the design of an optimized waverider. Figure 21 shows representative results for optimization using flow over a 1/2 power-law body at a Mach number of 6. The figure shows perspective views of the resulting waveriders when the different types of boundary layers are assumed. The value of L/D is 9.97 for the all laminar case, 7.35 for the all turbulent case, and 7.46 for the transiting case. The low value of L/D for the transiting flow suggests that much of the waverider surface is covered with turbulent flow. The data for this case confirms this conclusion, indicating an early transition from laminar to turbulent flow.

E. Computing Time

As stated earlier, 100 optimization levels or iterations are used in the optimization of a waverider configuration. This requires approximately 5 to 15 minutes of CPU time on a UNIVAC 1190 computer, depending on the type of flow field generating body used. The short running time of the waverider optimization program makes it possible to conduct parametric studies with these class of vehicles.

IV. CONCLUSIONS

The following conclusions are made from the present waverider optimization study:

1. The present results are the first examples of viscous optimized waveriders designed from axisymmetric flow fields, such as the hypersonic flow fields over minimum drag bodies.
2. The waveriders designed from flow fields over power-law bodies show promise, in terms of high L/D and minimum drag, when compared with conical-flow waveriders or other hypersonic configurations.
3. Optimization for a hypersonic cruise aircraft (maximizing L/D) results in a distinctly different configuration than optimization for a hypersonic accelerator aircraft (minimizing C_D).
4. The transition model used in the present study results in turbulent flow over much of the waverider surface. The nature of the boundary layer has a large effect on the aerodynamic characteristics of the vehicle. Better prediction methods for transition in hypersonic flows is badly needed.

V. ACKNOWLEDGEMENT

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REFERENCES

1. Anderson, John D., Jr., "A Survey of Modern Research in Hypersonics," AIAA Paper 84-1578, invited paper given at the AIAA 17th Fluid Dynamics, Plasma Dynamics, and Lasers Conference, Snowmass, Colorado, June 25-27, 1984.
2. Bowcutt, Kevin G., Anderson, John D., Jr., and Capriotti, D.P., "Viscous Optimized Hypersonic Waveriders," AIAA Paper 87-0272, presented at the AIAA 25th Aerospace Sciences Meeting, Reno, Nevada, January 12-15, 1987.
3. Bowcutt, K.G., Anderson, J.D., Jr., and Capriotti, D.P., "Numerical Optimization of Conical Flow Waveriders Including Detailed Viscous Effects," in Aerodynamics of Hypersonic Lifting Vehicles, AGARD-CP-428, March 1987, pp. 27/1-27/23.
4. Wallace Sawyer and Patrick Johnston, NASA Langley Research Center, private communication, 1985.
5. Bowcutt, Kevin G., Optimization of Hypersonic Waveriders Derived from Cone Flows--Including Viscous Effects, Ph.D. Dissertation, Department of Aerospace Engineering, University of Maryland, College Park, Maryland, 1986.

6. Corda, Stephen, Ph.D. Dissertation, Department of Aerospace Engineering, University of Maryland, College Park, Maryland, to be completed December 1987.
7. Nonweiler, T.R.F., "Aerodynamic Problems of Manned Space Vehicles," *Journal of the Royal Aeronautical Society*, Vol. 63, 1959, pp. 521-528.
8. Seddon, J. and Spence, A., "The Use of Known Flow Fields as an Approach to the Design of High Speed Aircraft," in *Hypersonic Boundary Layers and Flow Fields*, AGARD CP No. 30, May 1968, pp. 10/1-10/21.
9. Towend, L.H., "Research and Design for Lifting Reentry," *Progress in Aerospace Sciences*, Vol. 18, 1979, pp. 1-80.
10. Jones, J.G., Moore, K.C., Pike, J., and Roe, P.L., "A Method for Designing Lifting Configurations for High Supersonic Speeds, Using Axisymmetric Flow Fields," *Ingenieur-Archiv*, Vol. 37, pp. 56-72.
11. Rasmussen, M.L., "Waverider Configurations Derived from Inclined Circular and Elliptic Cones," *J. of Spacecraft and Rockets*, Vol. 17, No. 6, Nov.-Dec. 1980, pp. 537-545.
12. Rasmussen, M.L., and Clement, L.W., "Cone-Derived Waveriders with Longitudinal Curvature," AIAA Paper No. 84-2100, 1984.
13. Cole, J.D., and Zien, T.F., "A Class of Three-Dimensional Optimum Hypersonic Wings," *AIAA Journal*, Vol. 7, No. 2, Feb. 1969, pp. 264-271.
14. Kim, B.S., Rasmussen, M.L., and Jischke, M.C., "Optimization of Waverider Configurations Generated from Axisymmetric Conical Flows," AIAA Paper 82-1299, 1982.
15. Nelder, J.A., Mead, R., "A Simplex Method for Function Minimization," *Computer Journal*, Vol. 7, Jan. 1965, pp. 308-313.
16. Zucrow, M.J., and Hoffman, J.D., *Gas Dynamics*, John Wiley and Sons, New York, 1972.
17. Anderson, John D., Jr., *Modern Compressible Flow with Historical Perspective*, McGraw-Hill Book Co., New York, 1982.
18. Eckert, E.R.G., "Engineering Relations for Heat Transfer and Friction in High-Velocity Laminar and Turbulent Boundary-Layer Flow Over Surfaces With Constant Pressure and Temperature," *Transactions of the ASME*, Vol. 78, No. 6, August 1956.
19. DiCristina, V., "Three-Dimensional Laminar Boundary Transition on a Sharp 8 Degree Cone at Mach 10," *AIAA Journal*, Vol. 8, No. 5, May 1970, p. 855.
20. Van Wie, D.M., *An Application of Computational Fluid Dynamics to the Design of Optimum Ramjet and Powered Missile Components*, Ph.D. Dissertation, Department of Aerospace

Engineering, University of Maryland, College Park, Maryland, 1985.

21. Fetterman, D.E., "Favorable Interference Effects on Maximum Lift-Drag Ratios on Half-Cone Delta-Wing Configurations at Mach 6.86," NASA TN D-2942, August 1965.

TABLE 1

Experimental and Numerical Calculations of the Hypersonic Flow over a Cone-Wing

$M_\infty = 6.86$, $Re_L = 1.43 \times 10^6$, $L = 6.404$ in.
 $\theta_{\text{cone}} = 4^\circ$, $\Lambda_{\text{delta}} = 81^\circ$, $T_\infty = 59.2$ K, $T_0 = 617$ K

Case	$T_{\text{wall}}(\text{K})$	C_L	C_D	L/D
experiment	?	.012	.0045	2.67
numerical (freestream upper surface)	59.2 617.	.0126 .0126	.00329 .00294	3.83 4.29
numerical (1° compression upper surface)	59.2 617.	.0126 .0126	.00198 .00115	1.05 1.10

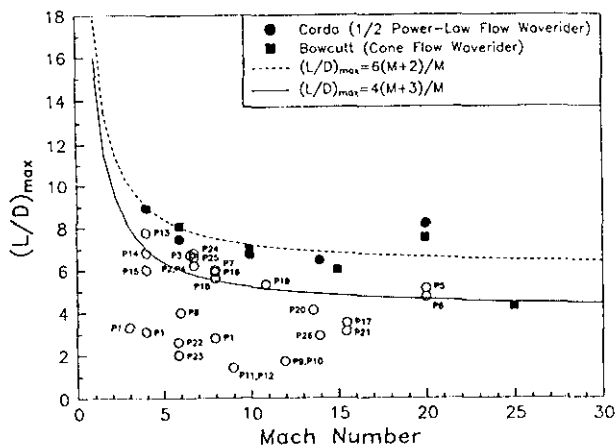
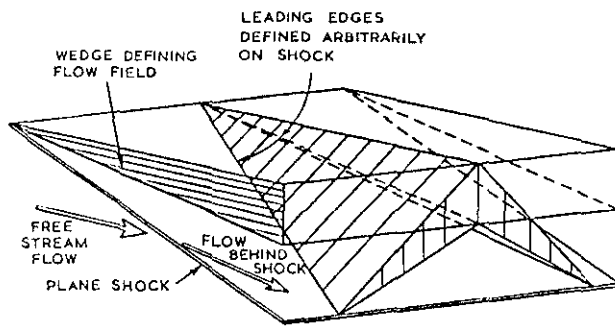
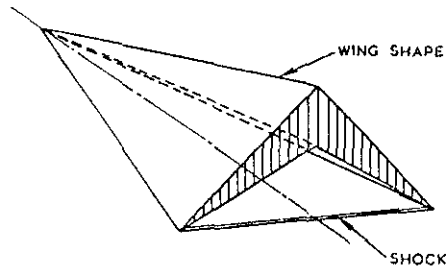


FIG. 1: Maximum lift-to-drag ratio comparison for various hypersonic configurations

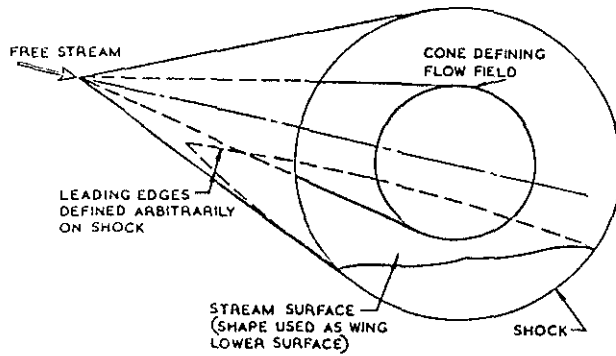


CONSTRUCTION FROM KNOWN FLOW FIELD

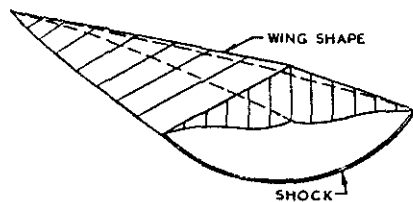


RESULTING WING AND SHOCK

FIG. 2: Nonweiler or caret wing (from Ref. 8)



CONSTRUCTION FROM KNOWN FLOW FIELD



RESULTING WING AND SHOCK

FIG. 3: Cone-flow wing (from Ref. 8)

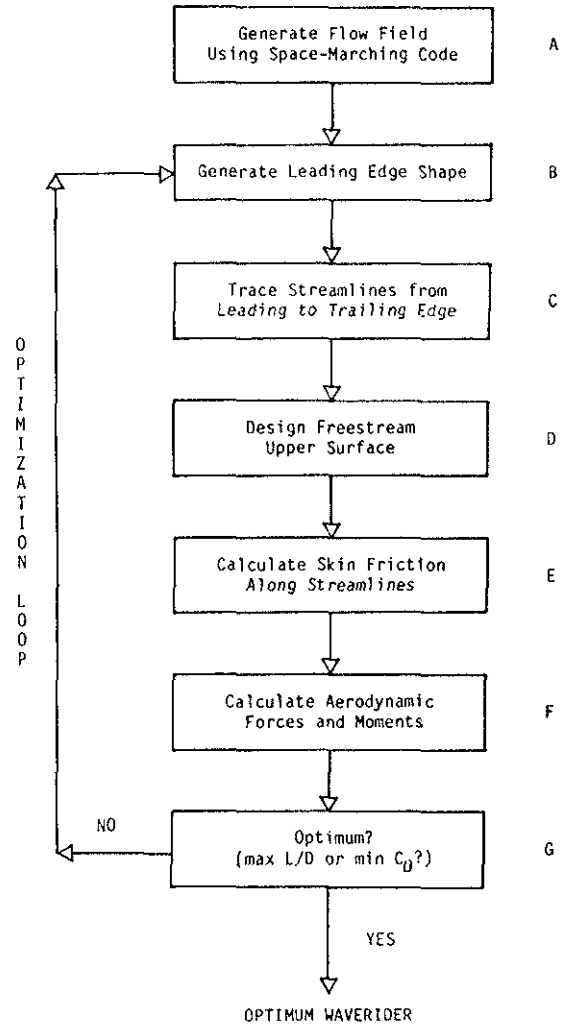


FIG. 4: Waverider design process

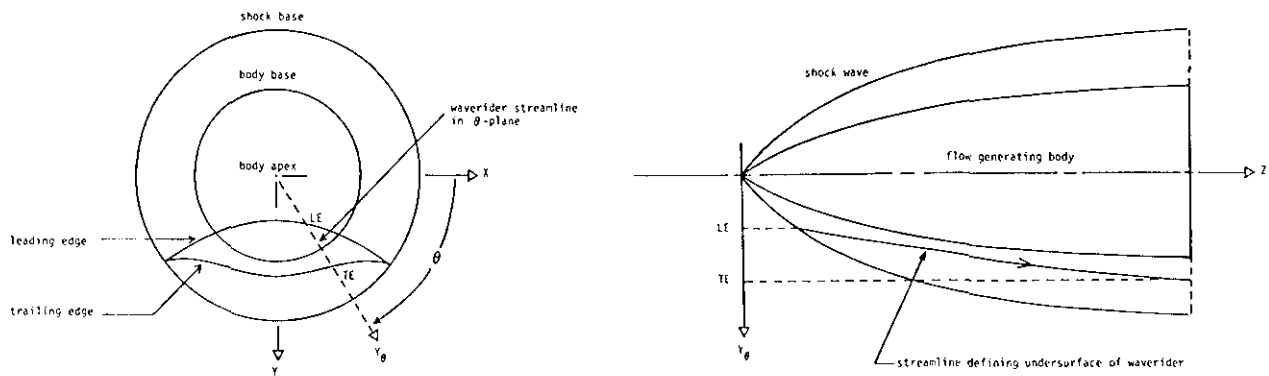


FIG. 5: Waverider construction

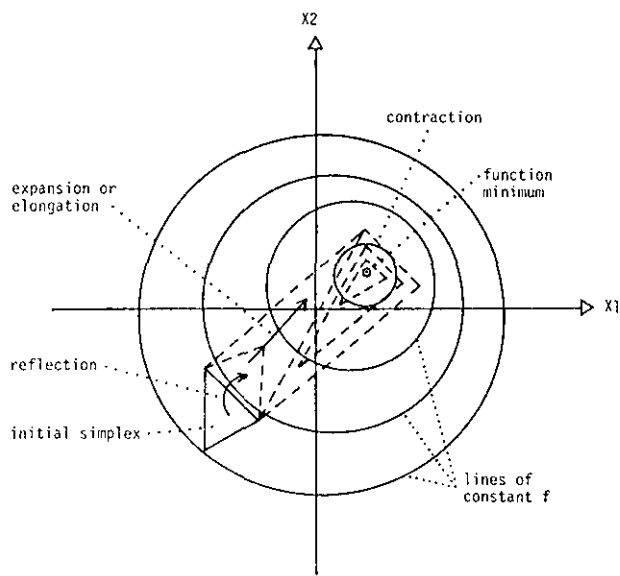


FIG. 6: Contour map example of simplex method for a function, f , of two variables

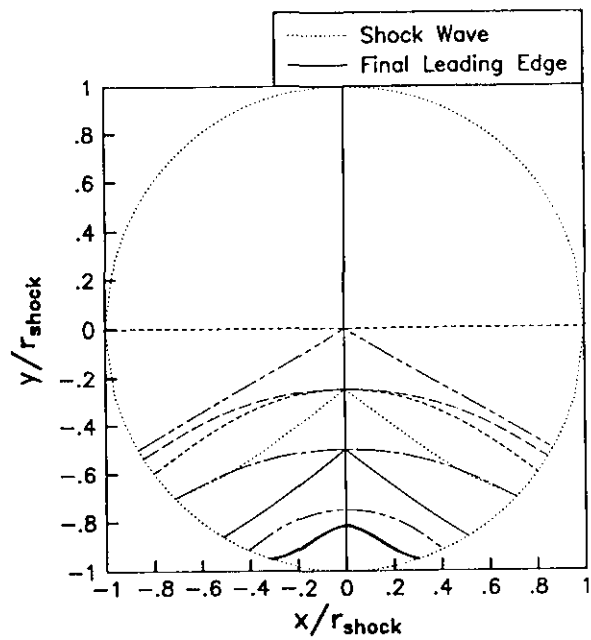


FIG. 7: Example of initial and optimized waverider design

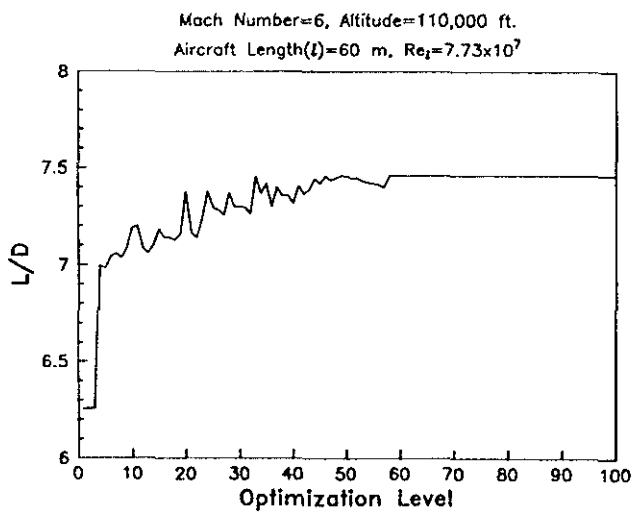


FIG. 8: Typical optimization history for waverider design

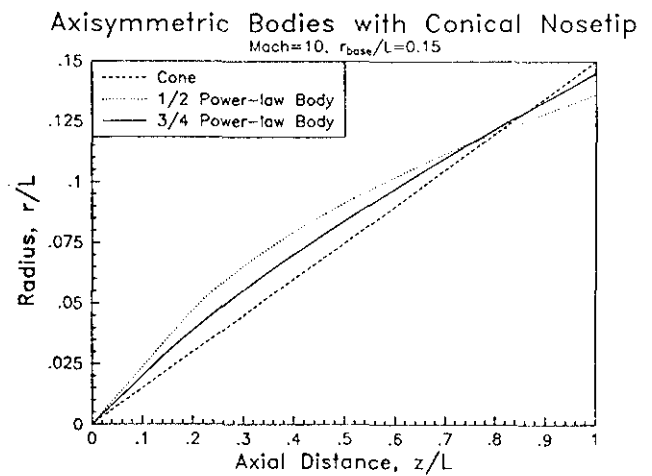


FIG. 9: Example of flow generating bodies

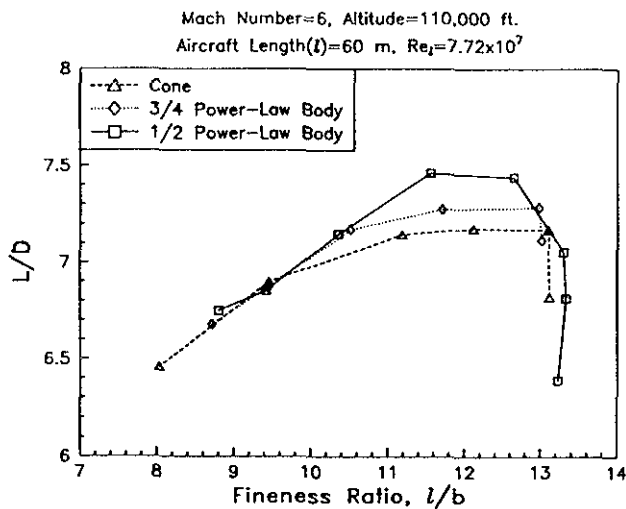


FIG. 10: Lift-to-drag ratio versus fineness ratio for a series of Mach 6 optimized waveriders

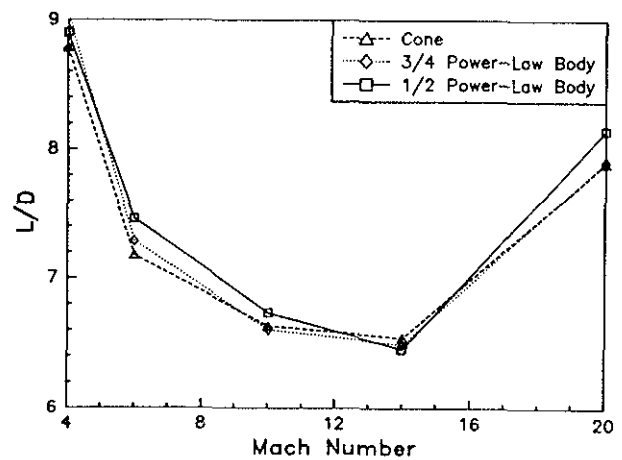


FIG. 11: Lift-to-drag ratio versus Mach number for the optimum waveriders

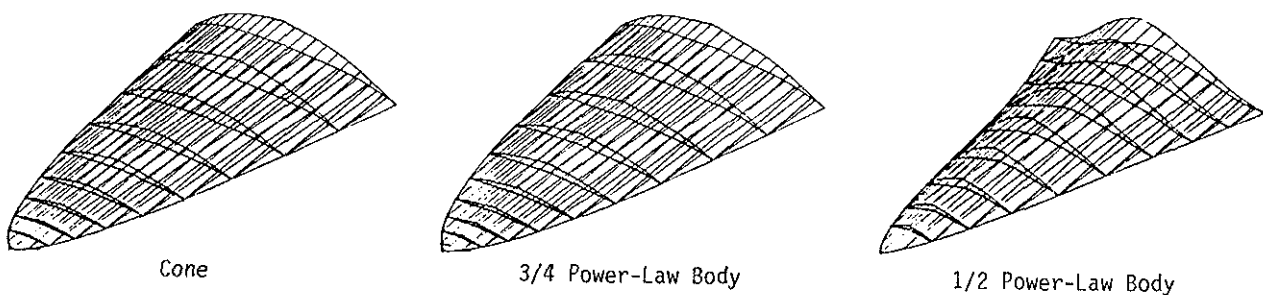


FIG. 12: Perspective views of Mach 6 optimum waveriders

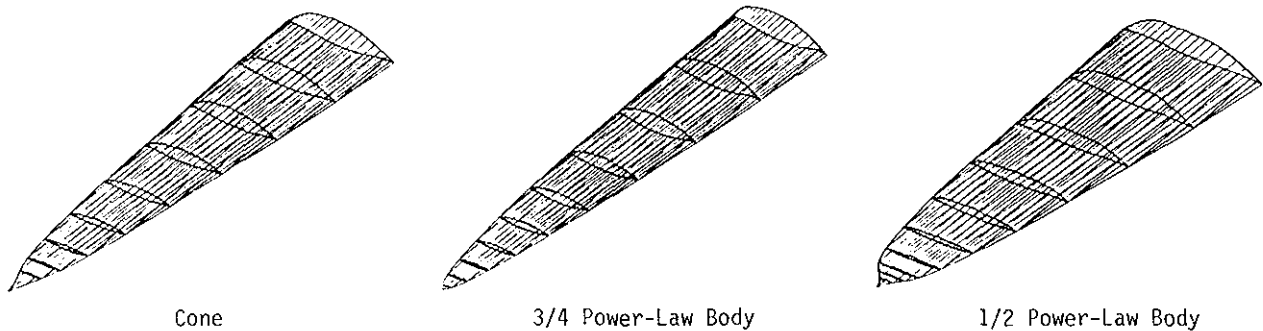


FIG. 13: Perspective views of Mach 20 optimum waveriders

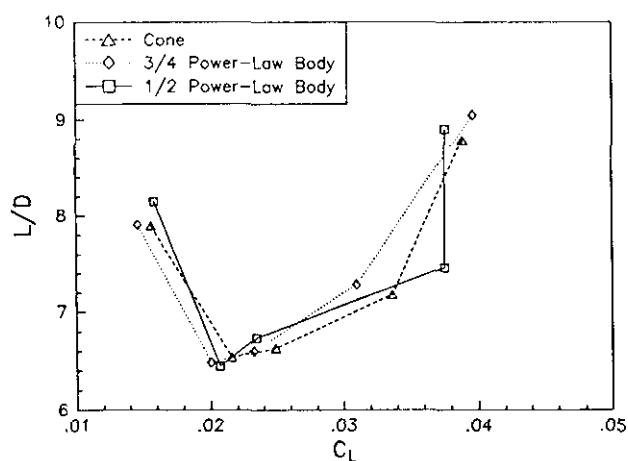


FIG. 14: Lift-to-drag ratio versus lift coefficient for the optimum waveriders

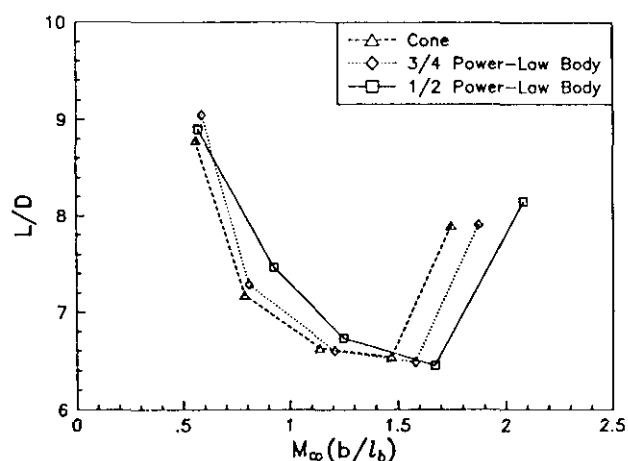


FIG. 15: Lift-to-drag ratio versus the hypersonic similarity parameter for the optimum waveriders

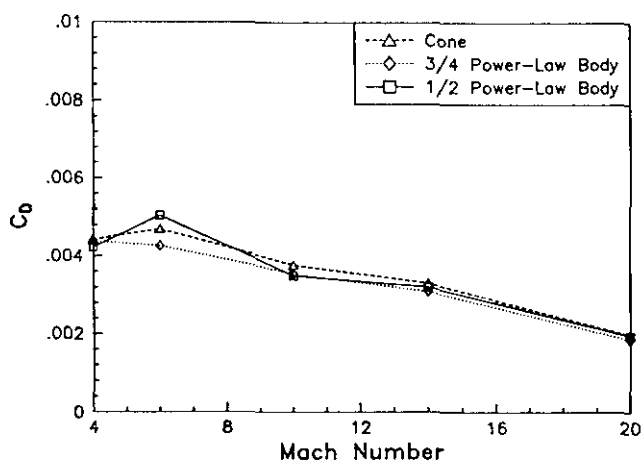


FIG. 16: Drag coefficient versus Mach number for the optimum waveriders

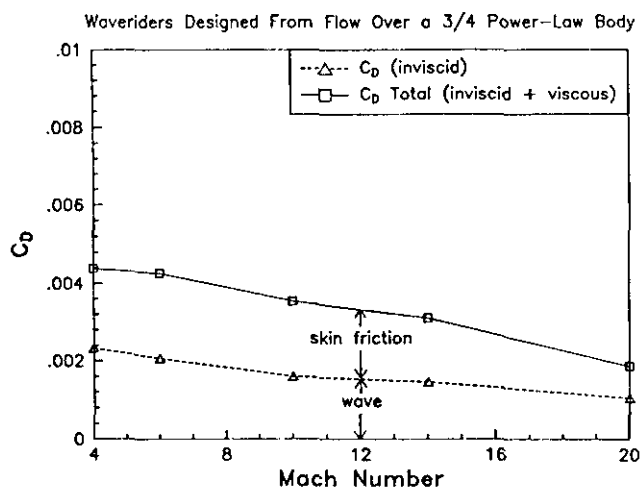


FIG. 17: Drag components for optimum waveriders designed from flow over a 3/4 power-law body

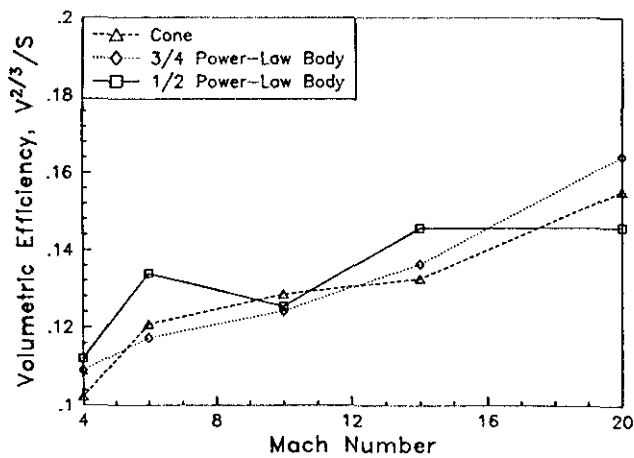


FIG. 18: Volumetric efficiency versus Mach number for the optimum waveriders

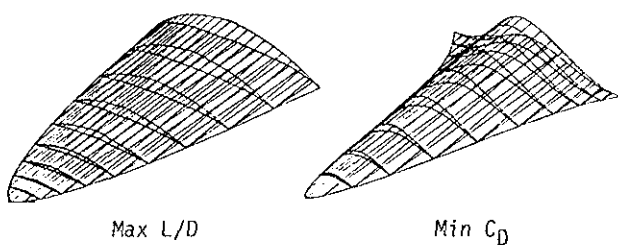


FIG. 19: Perspective views of Mach 6 waveriders optimized for different figures of merit

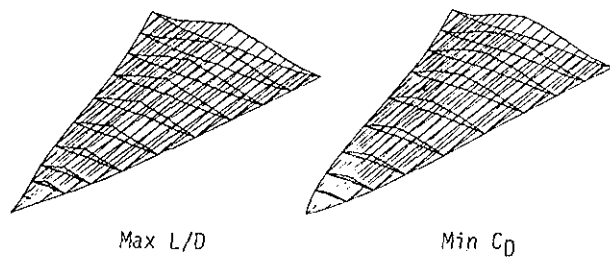


FIG. 20: Perspective views of Mach 14 waveriders optimized for different figures of merit

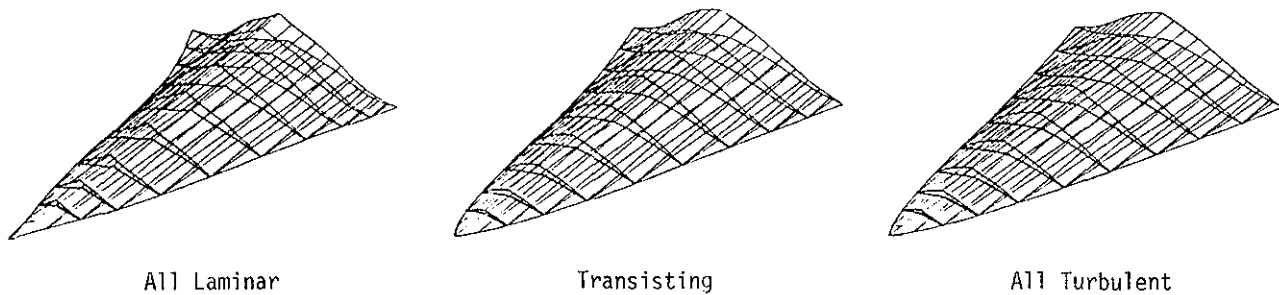


FIG. 21: Perspective views of optimized Mach 6 waveriders designed with different boundary layer characteristics